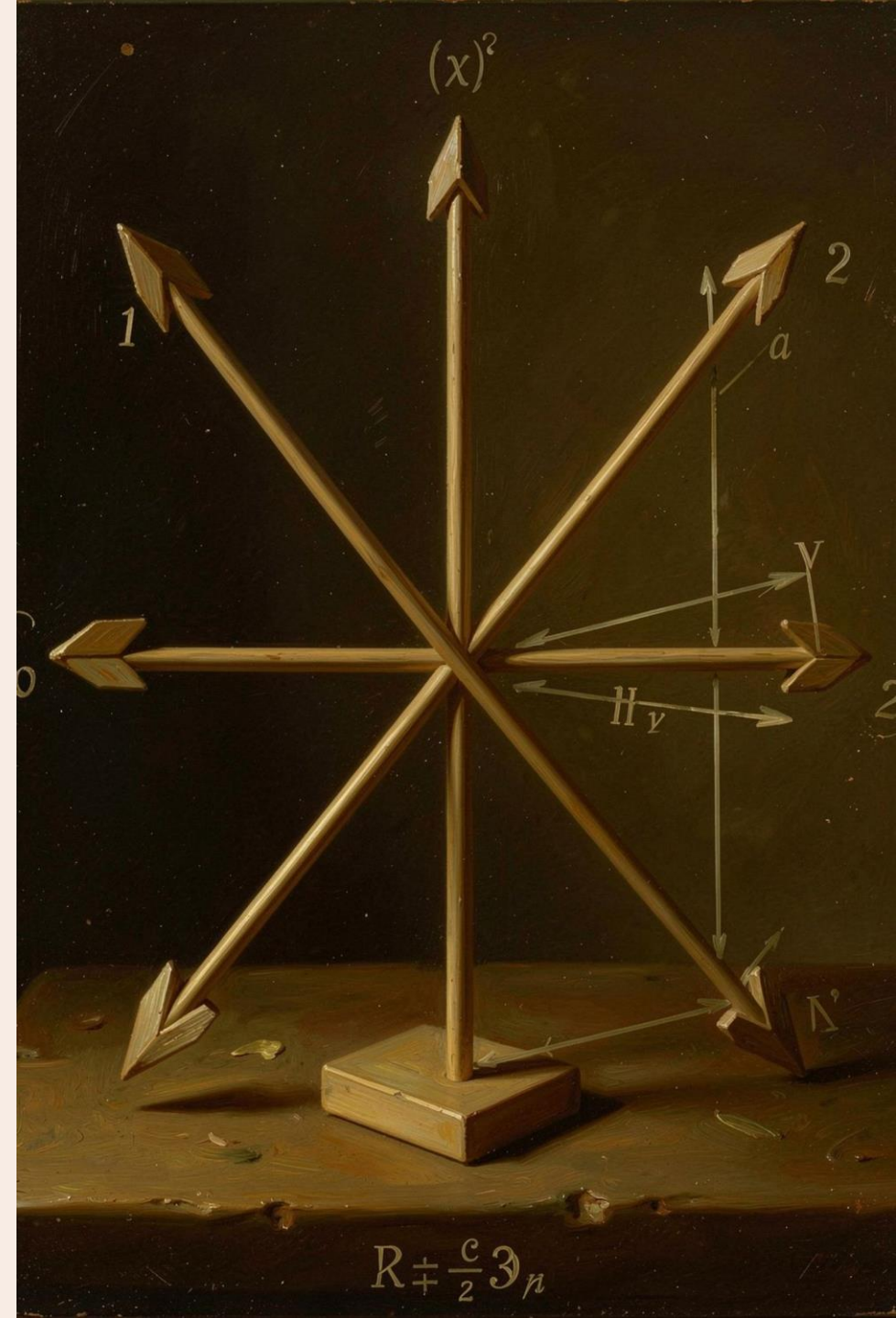


Vector Algebra

VECTOR ANALYSIS B.SC. ENGINEERING MATHEMATICS

A rigorous introduction to the algebraic and geometric operations that form the mathematical backbone of engineering mechanics, electromagnetics, fluid dynamics, and structural analysis. This module covers everything from Cartesian notation to triple products.



Lagrange’s Identity

Theorem: If $\underline{A}$ and $\underline{B}$ are two vectors and θ be the angle between them then $|\underline{A} \cdot \underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = |\underline{A}|^2 \cdot |\underline{B}|^2$

Proof: We know that, If $\underline{A}$ and $\underline{B}$ are two vectors and θ be the angle between them then

$$\underline{A} \cdot \underline{B} = |\underline{A}| |\underline{B}| \cos \theta \dots\dots\dots(i)$$

$$\& \quad \underline{A} \times \underline{B} = |\underline{A}| |\underline{B}| \sin \theta \dots\dots\dots(ii)$$

Squaring and Adding (i) & (ii) we get,

$$\begin{aligned} (\underline{A} \cdot \underline{B})^2 + (\underline{A} \times \underline{B})^2 &= |\underline{A}|^2 \cdot |\underline{B}|^2 \sin^2 \theta + |\underline{A}|^2 \cdot |\underline{B}|^2 \cos^2 \theta \\ &= |\underline{A}|^2 \cdot |\underline{B}|^2 (\sin^2 \theta + \cos^2 \theta) \end{aligned}$$

Or $(\underline{A} \cdot \underline{B})^2 + (\underline{A} \times \underline{B})^2 = |\underline{A}|^2 \cdot |\underline{B}|^2$ [Since $\sin^2 \theta + \cos^2 \theta = 1$]

$$\therefore |\underline{A} \cdot \underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = |\underline{A}|^2 \cdot |\underline{B}|^2$$

Since in vector analysis $(\underline{A} \cdot \underline{B})^2 = |\underline{A} \cdot \underline{B}|^2$ & $(\underline{A} \times \underline{B})^2 = |\underline{A} \times \underline{B}|^2$.

So, $|\underline{A} \cdot \underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = |\underline{A}|^2 \cdot |\underline{B}|^2$ **(Showed)**

Question: If $\underline{A} = \hat{i} + 3\hat{j} + \hat{k}$, $\underline{B} = 2\hat{i} + 4\hat{j} - 5\hat{k}$, prove that $|\underline{A} \cdot \underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = |\underline{A}|^2 \cdot |\underline{B}|^2$

Solution: Given, $\underline{A} = \hat{i} + 3\hat{j} + \hat{k}$, $\underline{B} = 2\hat{i} + 4\hat{j} - 5\hat{k}$,

$$\text{So, } |\underline{A}| = \sqrt{1^2 + 3^2 + 1^2} = \sqrt{1 + 9 + 1} = \sqrt{11}$$

$$\text{And } |\underline{B}| = \sqrt{2^2 + 4^2 + (-5)^2} = \sqrt{4 + 16 + 25} = \sqrt{45}.$$

$$\text{Also, } \underline{A} \cdot \underline{B} = (\hat{i} + 3\hat{j} + \hat{k}) \cdot (2\hat{i} + 4\hat{j} - 5\hat{k}) = 2 + 12 - 5 = 9$$

$$\text{And } \underline{A} \times \underline{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 1 \\ 2 & 4 & -5 \end{vmatrix} = (-15 - 4)\hat{i} - 5\hat{j} + (4 - 6)\hat{k} = -19\hat{i} + 7\hat{j} - 2\hat{k}$$

$$|\underline{A} \times \underline{B}| = \sqrt{(-19)^2 + 7^2 + (-2)^2} = \sqrt{414}$$

$$\text{L.H.S} = |\underline{A} \cdot \underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = 81 + 414 = 495 \text{ and } \text{R.H.S} = |\underline{A}|^2 \cdot |\underline{B}|^2 = 11 \cdot 45 = 495$$

$$\text{Hence, } |\underline{A} \cdot \underline{B}|^2 + |\underline{A} \times \underline{B}|^2 = |\underline{A}|^2 \cdot |\underline{B}|^2 \text{ (Proved)}$$

Vector differentiation:

Del/delta: A vector valued function is denoted and defined as follows:

$$\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

Gradient: If φ be a scalar valued function then the gradient of φ is denoted by $\text{grad } \varphi$ or $\underline{\nabla} \varphi$ and is defined as follows:

$$\text{grad } \varphi = \vec{\nabla} \varphi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \varphi = \frac{\partial \varphi}{\partial x} \hat{i} + \frac{\partial \varphi}{\partial y} \hat{j} + \frac{\partial \varphi}{\partial z} \hat{k}$$

Question: Suppose $\varphi = xyz$ then find the Gradient of φ at the point (2, 0, 1).

Solution: Given, $\varphi = xyz$

$$\text{So, } \mathbf{grad} \varphi = \vec{\nabla} \varphi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \varphi = \frac{\partial \varphi}{\partial x} \hat{i} + \frac{\partial \varphi}{\partial y} \hat{j} + \frac{\partial \varphi}{\partial z} \hat{k}$$

$$= \frac{\partial}{\partial x} xyz \hat{i} + \frac{\partial}{\partial y} xyz \hat{j} + \frac{\partial}{\partial z} (xyz) \hat{k}$$

$$= yz.1 \hat{i} + xz.1 \hat{j} + xy.1 \hat{k}$$

$$\vec{\nabla} \varphi = yz \hat{i} + xz \hat{j} + xy \hat{k}$$

At the point (2, 0, 1), $\text{grad } \varphi = \vec{\nabla} \varphi = 0 \cdot 1 \hat{i} + 2 \cdot 1 \hat{j} + 2 \cdot 0 \hat{k} = 2 \hat{j}$

Question: Suppose $\varphi = (xz - y^2)$ then find the Gradient of φ at the point (3, -1, 1).

Solution: $\text{grad } \varphi = \vec{\nabla} \varphi = \frac{\partial}{\partial x} (xz - y^2) \hat{i} + \frac{\partial}{\partial y} (xz - y^2) \hat{j} + \frac{\partial}{\partial z} (xz - y^2) \hat{k}$

$$= (z \cdot 1 - 0) \hat{i} + (0 - 2y) \hat{j} + (x \cdot 1 - 0) \hat{k}$$
$$= z \hat{i} - 2y \hat{j} + x \hat{k}$$

At the point (3, -1, 1), $\text{grad } \varphi = \underline{\nabla} \varphi = 1 \hat{i} - 2(-1) \hat{j} + 3 \hat{k} = \hat{i} + 2 \hat{j} + 3 \hat{k}$

Question: Suppose $\varphi = 2xz - yz$ then find the Gradient of φ at the point (2, -1, 0).

$$\text{grad } \varphi = \vec{\nabla} \varphi = \frac{\partial}{\partial x} (2xz - yz) \hat{i} + \frac{\partial}{\partial y} (2xz - yz) \hat{j} + \frac{\partial}{\partial z} (2xz - yz) \hat{k}$$
$$= (2z \cdot 1 - 0) \hat{i} + (0 - z \cdot 1) \hat{j} + (2x \cdot 1 - y \cdot 1) \hat{k}$$
$$= 2z \hat{i} - z \hat{j} + (2x - y) \hat{k}$$

At the point (2, -1, 0), $\text{grad } \varphi = \vec{\nabla} \varphi = 2 \cdot 0 \hat{i} - 0 \hat{j} + (2 \cdot 2 + 1) \hat{k} = 5 \hat{k}$

Divergence: If $\vec{A} = f\hat{i} + g\hat{j} + h\hat{k}$ be a vector valued function then the divergence of $\vec{A}$ is denoted by $\text{div } \vec{A}$ or $\vec{\nabla} \cdot \vec{A}$ and is defined as follows:

$$\text{div } \underline{A} \text{ or } \vec{\nabla} \cdot \vec{A} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (f\hat{i} + g\hat{j} + h\hat{k}) = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$$

Question-1: If $\vec{A} = xz \hat{i} - xy \hat{j} + yz \hat{k}$ then find the Divergence of $\vec{A}$ at the point (1,-1, 2).

Solution: Given, $\vec{A} = xz \hat{i} - xy \hat{j} + yz \hat{k}$

$$\begin{aligned} \text{Now, } \text{div } \vec{A} \text{ or } \vec{\nabla} \cdot \vec{A} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (xz \hat{i} - xy \hat{j} + yz \hat{k}) \\ &= \frac{\partial}{\partial x} (xz) - \frac{\partial}{\partial y} (xy) + \frac{\partial}{\partial z} (yz) = z - x + y \end{aligned}$$

At the point (1,-1, 2), $\text{div } \underline{A} \text{ or } \vec{\nabla} \cdot \vec{A} = 2 - 1 - 1 = 0$

Question-2: If $\vec{A} = x^2\hat{i} - z^2\hat{j} - y^2z\hat{k}$ then find the Divergence of $\underline{A}$ at the point $(-3, 0, 2)$.

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(z^2) - \frac{\partial}{\partial z}(y^2z) = 2x - 0 - y^2 = 2x - y^2$$

At the point $(-3, 0, 2)$, $\text{div } \vec{A}$ or $\vec{\nabla} \cdot \vec{A} = 2(-3) - 0 = -6$

Exercise: Let $\vec{P} = x^2yz\hat{i} - (xz - y^2)\hat{j} - yz\hat{k}$, find the Divergence of $\vec{P}$ at the point $(2, 1, -3)$.

Curl: If $\vec{A} = f\hat{i} + g\hat{j} + h\hat{k}$ be a vector valued function then the curl of $\vec{A}$ is denoted by $\text{curl } \vec{A}$ or $\vec{\nabla} \times \vec{A}$ and is defined as follows:

$$\text{curl } \vec{A} = \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix} = \hat{i} \left(\frac{\partial h}{\partial y} - \frac{\partial g}{\partial z} \right) - \left(\frac{\partial h}{\partial x} - \frac{\partial f}{\partial z} \right) \hat{j} + \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) \hat{k}$$

Question: If $\vec{A} = x^2\hat{i} - z^2\hat{j} + y^2\hat{k}$ then find the Curl of $\vec{A}$ at the point (1,0, 1).

Solution: Given, $\vec{A} = x^2\hat{i} - z^2\hat{j} + y^2\hat{k}$.

$$\begin{aligned}\vec{\nabla} \times \vec{A} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & -z^2 & y^2 \end{vmatrix} \\ &= \left(\frac{\partial}{\partial y} y^2 + \frac{\partial}{\partial z} z^2 \right) \hat{i} - \left(\frac{\partial}{\partial x} y^2 - \frac{\partial}{\partial z} x^2 \right) \hat{j} + \left(-\frac{\partial}{\partial x} z^2 - \frac{\partial}{\partial y} x^2 \right) \hat{k} \\ &= (2y + 2z)\hat{i} - (0 - 0)\hat{j} + (0 - 0)\hat{k} \\ &= (2y + 2z)\hat{i}\end{aligned}$$

At the point (1,0, 1), $\vec{A}$ or $\vec{\nabla} \times \vec{A} = (2.0 + 2.1)\hat{i} = 2\hat{i}$